



# ISLAMIC UNIVERSITY OF TECHNOLOGY

## Course Outline and Course Plan



|                                      |                                                                                         |                            |                                                            |
|--------------------------------------|-----------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------|
| <b>Name of the Teacher</b>           | Shahriar Ivan                                                                           | <b>Position</b>            | Assistant Professor                                        |
| <b>Department</b>                    | CSE                                                                                     | <b>Program</b>             | B.Sc. Engineering in SWE                                   |
| <b>Course Code</b>                   | CSE 4107                                                                                | <b>Course Title</b>        | Structured Programming I                                   |
| <b>Academic Year</b>                 | 2024-25                                                                                 | <b>Semester</b>            | Winter                                                     |
| <b>Contact Hours</b>                 | 3.00                                                                                    | <b>Credit Hours</b>        | 3.00                                                       |
| <b>Textbooks and Reference books</b> | 1. C Programming - A Modern Approach<br>2. Teach Yourself C<br>3. Programming in ANSI C | <b>Author of the Books</b> | 1. K. N . King<br>2. Herbert Schildt<br>3. E. Balagurusamy |
| <b>Prerequisites</b>                 | A good grasp on College Algebra and Problem Solving                                     |                            |                                                            |
| <b>Corequisites</b>                  |                                                                                         |                            |                                                            |
| <b>Course Homepage</b>               | Google Classroom Code: czyfcbci                                                         |                            |                                                            |
| <b>Teaching Methods/ Approaches</b>  | Lecture<br>Demonstration<br>Problem Solving                                             |                            |                                                            |
| <b>Teaching Aids</b>                 | Multimedia and OHP<br>Board and Marker<br>Video Conferencing Tool and Graphics Tablet   |                            |                                                            |

### Course Assessment Method

| Attendance (0%) | Quiz/Viva (20%) (Best 3 out of 4) |                      |                      |                      | Mid Exam (40%) | Final Exam (40%) |
|-----------------|-----------------------------------|----------------------|----------------------|----------------------|----------------|------------------|
|                 | 1 <sup>st</sup> Quiz              | 2 <sup>nd</sup> Quiz | 3 <sup>rd</sup> Quiz | 4 <sup>th</sup> Quiz | Week/ Date     | Week/ Date       |

|  |                      |                      |                       |                       |                        |                          |
|--|----------------------|----------------------|-----------------------|-----------------------|------------------------|--------------------------|
|  | Week/<br>Date        | Week/<br>Date        | Week/<br>Date         | Week/<br>Date         |                        |                          |
|  | 4 <sup>th</sup> Week | 7 <sup>th</sup> Week | 10 <sup>th</sup> Week | 14 <sup>th</sup> Week | 8-9 <sup>th</sup> Week | 17-18 <sup>th</sup> Week |

| Grading Policy   |              |             |                  |              |             |
|------------------|--------------|-------------|------------------|--------------|-------------|
| Marks out of 100 | Letter Grade | Grade Point | Marks out of 100 | Letter Grade | Grade Point |
| 80 - 100         | A+           | 4.00        | 55 - 59          | B-           | 2.75        |
| 75 - 79          | A            | 3.75        | 50 - 54          | C+           | 2.50        |
| 70 - 74          | A-           | 3.50        | 45 - 49          | C            | 2.25        |
| 65 - 69          | B+           | 3.25        | 40 - 44          | D            | 2.00        |
| 60 - 64          | B            | 3.00        | 0 - 39           | F            | 0.00        |

| Course Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Intro to C Programming Language:</b> Introduction to programming languages, Standards, Basic I/O</p> <p><b>Programming Style:</b> Formal aspects of language syntax and semantics</p> <p><b>Basic and Advanced Language Features:</b> Operators, Conditional Statements, Loops, Functions, File I/O etc.</p> <p><b>Problems Solving Techniques:</b> Define the problem, List possible solutions, Evaluate methods, Create an implementation plan, Communicate the solution</p> <p><b>Programming Environments and Tools:</b> Use of different programming development environments and tools</p> |

| Course Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>The course aims to provide the student with:</p> <ol style="list-style-type: none"> <li>1. Knowing the basics of Computer Programming Language</li> <li>2. Exploring standard I/O, Conditional statements, Iteration statements</li> <li>3. Learning about Data types and Variables</li> <li>4. Using Arrays and Functions in different scenarios</li> <li>5. Understanding Pointers and Dynamic Memory Allocation</li> <li>6. Acquiring knowledge about Structures and Unions</li> <li>7. Getting insights on File I/O and Low-Level Programming</li> </ol> |

| Mapping with CO, PO, and Bloom's Taxonomy |                                                  |                            |                                    |
|-------------------------------------------|--------------------------------------------------|----------------------------|------------------------------------|
| CO No.                                    | Course Outcomes (CO) Statement                   | Levels of Bloom's Taxonomy | Matching with Program Outcome (PO) |
| CO1                                       | Identify different programming language concepts | C1 (Remember)              | PO1                                |

|     |                                                                                 |                 |          |
|-----|---------------------------------------------------------------------------------|-----------------|----------|
| CO2 | Explain which application scenario requires which Programming language features | C2 (Understand) | PO1      |
| CO3 | Apply the programming language expertise to solve complex engineering problems  | C3 (Apply)      | PO1, PO2 |

| Weekly Plan for Course Content and Mapping with CO |                                                                                                                                                                                        |               |  |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--|
| Weeks                                              | Topics                                                                                                                                                                                 | COs           |  |
| 1                                                  | Introduction, Origin, Standards, Comments, Basic I/O, Identifiers, Keywords                                                                                                            | CO1           |  |
| 2                                                  | Fundamentals of C                                                                                                                                                                      | CO1           |  |
| 3                                                  | The Printf Function, The Scanf Function                                                                                                                                                | CO1           |  |
| 4                                                  | Arithmetic Operators, Assignment Operator, Increment and Decrement Operators, Expression Evaluation                                                                                    | CO1, CO2      |  |
| 5                                                  | Logical Expressions, The If Statement, The Switch Statement                                                                                                                            | CO1, CO2      |  |
| 6                                                  | The While Statement, The Do Statement, The For Statement, Exiting from a Loop, The Null Statement                                                                                      | CO1, CO2      |  |
| 7                                                  | Integer Types, Float Types, Character Types, Type Conversion                                                                                                                           | CO1           |  |
| 8                                                  | Mid Exam                                                                                                                                                                               |               |  |
| 9                                                  |                                                                                                                                                                                        |               |  |
| 10                                                 | One-Dimensional Arrays, Multidimensional Arrays, Variable-Length Arrays, Function Definition, Calling Functions, Declarations, Arguments, The return Statement, Termination, Recursion | CO1, CO2, CO3 |  |
| 11                                                 | Pointer Variables, Assignment, Pointers as Argument, Pointers as Return values                                                                                                         | CO1, CO2, CO3 |  |
| 12                                                 | Pointer Arithmetic, Using Pointers for Array Processing, Using Array name as Pointer, Pointers and Multidimensional Arrays                                                             | CO1, CO2, CO3 |  |
| 13                                                 | String Literals, String Variables, Reading and Writing Strings, Accessing Characters in a String, C String Library, Arrays of Strings                                                  | CO1, CO2, CO3 |  |
| 14                                                 | Structure Variables, Structure Types, Nested Arrays and Structures, Unions, Enumerations, Streams, File Operations, File I/O, File Positioning                                         | CO1, CO2, CO3 |  |
| 15                                                 | Dynamic Storage Allocation, Dynamically Allocated Strings, Dynamically Allocated Arrays, Deallocating Storage, Linked List, Pointers to Pointers                                       | CO1, CO2, CO3 |  |
| 16                                                 | How Preprocessor works, Preprocessor Directives, Macro Definitions, Conditional Compilation, Bitwise Operations, Bit-Fields in Structures                                              | CO1, CO2, CO3 |  |
| 17                                                 | Final Exam                                                                                                                                                                             |               |  |

|    |  |  |
|----|--|--|
| 18 |  |  |
| 19 |  |  |

| Program Outcomes |                                                                                                                                                                                                                                                                                             |  |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| PO 1             | <b>Engineering Knowledge:</b> Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.                                                                                                      |  |
| PO 2             | <b>Problem Analysis:</b> Identify, formulate, research and analyze complex engineering problems and reach substantiated conclusions using the principles of mathematics, the natural sciences and the engineering sciences.                                                                 |  |
| PO 3             | <b>Design/Development of Solutions:</b> Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety and of cultural, societal and environmental concerns.         |  |
| PO 4             | <b>Investigation:</b> Conduct investigations of complex problems, considering experimental design, data analysis and interpretation and information synthesis to provide valid conclusions.                                                                                                 |  |
| PO 5             | <b>Modern Tool Usage:</b> Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modeling, to complex engineering activities with an understanding of their limitations.                                                  |  |
| PO 6             | <b>The Engineer and Society:</b> Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.                                                          |  |
| PO 7             | <b>Environment and Sustainability:</b> Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of and need for sustainable development.                                                                            |  |
| PO 8             | <b>Ethics:</b> Apply ethical principles and commit to the professional ethics, responsibilities and the norms of the engineering practice.                                                                                                                                                  |  |
| PO 9             | <b>Individual Work and Teamwork:</b> Function effectively as an individual and as a member or leader of diverse teams and in multidisciplinary settings.                                                                                                                                    |  |
| PO 10            | <b>Communication:</b> Communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions. |  |
| PO 11            | <b>Project Management and Finance:</b> Demonstrate knowledge and understanding of engineering and management principles and apply these to one's work as a team member or a leader to manage projects in multidisciplinary environments.                                                    |  |
| PO 12            | <b>Life-Long Learning:</b> Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.                                                                                                        |  |

| Mapping of COs and POs [Correlation level 1 for low, 2 for moderate and 3 for high] |      |      |      |      |      |      |      |      |      |       |       |       |
|-------------------------------------------------------------------------------------|------|------|------|------|------|------|------|------|------|-------|-------|-------|
| Course Outcomes                                                                     | PO 1 | PO 2 | PO 3 | PO 4 | PO 5 | PO 6 | PO 7 | PO 8 | PO 9 | PO 10 | PO 11 | PO 12 |

|      |   |   |  |  |  |  |  |  |  |  |  |  |
|------|---|---|--|--|--|--|--|--|--|--|--|--|
| CO 1 | 3 |   |  |  |  |  |  |  |  |  |  |  |
| CO 2 | 2 |   |  |  |  |  |  |  |  |  |  |  |
| CO 3 | 2 | 3 |  |  |  |  |  |  |  |  |  |  |

**Student Consulting Hour:**

Thursday, 10 a.m. - 12 p.m.

Saturday, 10 a.m. - 11 p.m. (on request)

**Instruction Contact Details:**

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